NyLaw: Dataset for Georgia's Civil Pro Se Court Cases

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by
Nyla Lawrence

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Abstract

The legal process in the United States is extensive and prolonged. Artificial intelligence has already made its way through parts of the legal system, posing questions, and ethical and technical usage such as privacy concerns and reference hallucinations. In conjunction with the rise in artificial intelligence implemented into day-to-day living and work decisions, understanding the intersectionality between the two fields is critical. Instead of ignoring the inevitable higher integration of this technology, it is paramount that work is done to properly train and utilize these new systems for a more efficient and accurate legal process. To use machine learning requires the processing of thousands to millions of data points to accurately learn and predict or classify. Currently, there is not a concise dataset of civil state court cases, and especially none for pro se cases, that can train machine learning algorithms in the United States. This project curates one of these datasets of legal cases as a baseline, and quantifies them for immediate use within machine learning algorithms. As a pro se filer, this project could severely simplify the work done and the process to traverse their legal battle. It is constrained to one state to simplify the problem but can be simulated across states and jurisdictions for future works. Alongside being my home state, Georgia is currently evaluating this same intersectionality within judicial committees, making it the choice of state to be in line with current research on the topic.

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Dedication

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I wish for the best for Kya and Malachi as I also watch them grow into their own persons.

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1
Introduction

1.1 Legalese

1.1.1 Court and Case Structure across the United States

The federal court system in the United States is broken up from highest to lowest hierarchy via the following structure: Supreme Court, Court of Appeals, District Courts, Bankruptcy Courts, and Article I Courts via Article 3 of the United States Constitution (Library of Congress n.d.) and Title 28 of the United States Code (United States Congress 2017). Article 1 courts include appellate courts for veteran claims, active military cases, and tax court Judicial Council of Georgia n.d. Cases usually start with state courts and grow to the appellate and federal level depending on the case and specific violation or actions being investigated. Each state has their own set of courts and structure on top of this. Specific courts focus on a particular type of conduct which include civil, criminal, appellate, and bankruptcy. Georgia specifically holds the following precedence and structure: Supreme Court, Court of Appeals, Superior Court, in order of precedence [1], followed by statewide business court, state court, juvenile court, probate court, magistrate court, and municipal court (Judicial Council of Georgia 2022). The state courts, Superior courts, and magistrate courts hear and try civil cases based on the severity and type

[1] The courts listed are in order of power and spread of judgment. Judgments released by the Supreme Court hold legal principles and ideology that lower courts must follow. On the other hand, state Superior Courts judgments do not hold power over the Court of Appeals or any of the other courts.

of civil case presented. The focus of this paper is around these three courts and the civil cases that they hear and try. These cases include wage garnishment, auto abandonment, small claims [2], medical malpractice, distress warrants [3], divorce, child support, and others. Appellate and federal court cases are outside the scope of the current research project. It is encouraged that future research expand into these areas and understand the intricacies of civil cases through these secondary and higher channels.

1.2 Financial Responsibilities

At any level, it is customary for both or all parties to obtain a lawyer. To do so, one must pay a rolling fee called a retainer, to secure the lawyer's attention and resources for a given amount of time. Resources such as travel time are also included in these billable hours and are all calculated by the lawyer or law firm on an hourly basis. After an initial retainer has been paid, then the lawyer is officially assigned and deemed as working the case. Depending on the type of cases, the risk management involved, and the prominence of the lawyer and firm, the retainer may be higher or lower. In Georgia, the hourly rate was between $200 and $350 per hour in 2023 (World Population Review 2025). Assuming a general 9:00 AM to 5:00 PM schedule and managing about 5 cases at a time, in one month a retainer would cost the client about $8,800 [4]. To maintain their working relationship, they must reinstate the account by paying another retainer.

Given these base figures, it is important to calculate the average time a civil court case lasts to understand the depth of financial responsibility of the party. Overall costs depend on the length of the case, staying or changing lawyers, plus possible inflation. Generally, a case can take anywhere from a few months to years spanning the initial contact and entering the court system to exiting the court system. This does not include appeals or changes in jurisdiction,

[2] Small claims are lower monetary requests from one party to another. They range from state to state but generally are charges no higher than $15,000.

[3] Distress warrants are warrants for the debtor to reclaim items which were not paid for by the other party in order to settle their debt.

[4] Assuming a strict 40 hour work week and working 8 hours a week on the case. This is multiplied by the average Georgia billable hours to attain this value of $8,800 per month.

that would add additional time and processing of a case. Georgia has a recorded range from 4 months to 4 or more years for a personal injury case to be completed, which depends on the severity and complexities of the case (Gordon 2025). With the various amounts of work done, a "simple" personal injury case could cost the client well over $15,000 estimated value for the four months of work done by the lawyer and the various court fees. This figure only increases as the case continues in duration and various fees present themselves.

1.2.1 Pro Se Definition, Rights, and Responsibilities

As stated, it is customary to obtain a lawyer as they ensure a level of expertise and understanding of the legal system. However, this is not a mandatory requirement for a case to proceed in the courts. Pro se is the Latin word meaning "for oneself" (Legal Information Institute, Cornell Law School 2025). This has been applied to law in the case where a person decides to not utilize a lawyer in court to defend them. Instead, they defend themselves and uphold all of the legal communication and duties that a regular lawyer would do. This is an option for many people due to the financial burden of hiring and retaining a lawyer, having previous experience in law, or for other reasons. A client can also fire their lawyer at any point in the case and choose to proceed pro se. In a trade off for not paying a lawyer, pro se filers usually bring their lack of expertise and legal experience into the case which adds a time component to their personal lives by researching, filing documents, and communicating the case. This option is one that many cases take despite the trade off. Filing pro se was a significant portion of civil cases, claiming 27% percent of cases in 2001 trends (Administrative Office of the U.S. Courts 2021a). Of these, 69% of pro se cases are filed by incarcerated individuals (Administrative Office of the U.S. Courts 2021a).

The US Judiciary Act has a section called "the Code of Conduct for United States Judges, the Federal Rules of Civil Procedure, the Federal Rules of Criminal Procedure, the Federal Rules of Evidence, the Federal Rules of Appellate Procedure, and the Federal Rules of Bankruptcy Procedure" describing the rights and provisions of pro se filer within civil and criminal courts

Administrative Office of the U.S. Courts 2021b. Courts, and specifically the Georgia courts, have offices and special resources for pro se case filers to ensure basic access to legal help and tips for case management. There are cases and special rules when both the plaintiff and the defendant represent themselves pro se. Although declaring pro se does eliminate the lawyer fees, it does not eliminate all of the costs associated with the case. There still exist several fees which include but are not limited to filing fees, registration of judgement, and miscellaneous fees such as record searching, reproducing an electronic format of records, per page reproduction of records, and moreUnited States District Court, Northern District of Georgia 2025. Although there are services within court systems that assist pro se filers, there are still disparities in financial responsibilities, access to resources, time spent understanding legal structures, and representing yourself in court that have yet to be widely addressed. This project aims to build the dataset necessary for a future development that would assist these pro se filers by equipping them with a case optimization application powered by machine learning. To do so, this project attempts to curate the first publicly available baseline for machine learning algorithms on civil pro se cases in Georgia. By curating this dataset, the next goal would be to create a machine learning model that can understand strategies that worked between each case and incorporate the laws of the courts in present time. This learning system would leverage case specific guidance in concatenation with current legal structures to provide better assistance with a pro se filer's case. The time used to research cases and understand the next steps they underwent could significantly change the way pro se cases are conducted and the success rate of cases themselves.

1.3 Current State of Machine Learning in Law

In the new age of artificial intelligence and machine learning, systems are built to make tasks more efficient. To build such systems, the models need large amounts of data to be trained to learn patterns, and to develop results which are just as intricate. This data must be large enough and have a depth which can encompass the types of problems or requests that the model will encounter after launch. Although current research focuses heavily on the optimization and

usage of artificial systems, it is important to focus on the data which drives such systems. The models learn from the datasets and not everyday interactions, which limits the scope of understanding to simply what is presented in the data. Having underlying biases, missing values, lack of data, and neglecting underrepresented data curates models that reproduce these same inconsistencies and incorrect interpretations. Even with an equitable and ethical dataset, generative artificial intelligence is prone to hallucinations and making its way into the court systems. An example is the Texas case of James Gauthier vs. GoodYear Tire & Robber Co., where the Plaintiff's Lawyer, Brandon Monk, utilized ChatGPT to cite in a legal document supporting court cases which turned out to be fictitious Service 2024. The lawyer was fined $2,000 for this incident Tsipursky 2025. A similar instance occurred in New York with Roberto Mata vs. Avianca, Inc. and the lawyers being fined $5,000 Archive 2025. There are even separate forums, like Incidientdatabase.ai, which host hallucinations and unprecedented behavior of machines in public spaces such as the ones fined in the court cases.

Machine learning in the legal field has continued to grow over the years and throughout many countries. The European Union employed the Offender Assessment System, also called OASys, which is an AI program that helps predict the likelihood to re-offend, which is packaged under the term "recidivism". It is currently being used within England and Wales legal systems to assist in decision making on sentencing length and parole possibilities (Tiarks 2023). With many inter-correlated parts of the legal system, mental stability, and previous history, regular methods of prediction become less effective at making predictions. To reach the stage it is currently in however, the model was trained on slightly less than 850,000 offender assessments. Systems such as this are starting to grow and be more integrated into the legal system. Machine learning is already in use for document review, case analysis, and case predictions (Kerdvibulvech 2024). These practices are expanding the technical areas of Name Entity Recognition, document automation, relevant document retrieval, and classification (Siino et al. 2025) However, most of these systems are at the hands of large entities such as law firms and judges, with few free

applications or sources of data being available for civilians without a human lawyer, let alone an legal backed AI assist.

2
Literature Review

2.1 History of Datasets

2.1.1 Past

With companies utilizing artificial intelligence models specified for legal aid, such as LawConnect, and general models, such as ChatGPT, it is important to build better data to power these models and understand how these models came about in the first place. Alan M. Turing created the beginning of what we now consider artificial intelligence by posing the simple question, "Can machines think?" (Turing 1950). The inception of artificial intelligence as a formal field came from John McCarthy's 1956 workshop (McCarthy 1959) in which he proposed computers could mimic human decision making via logical programming, but would require more studying into how the human brain worked before we could instruct computers. Arthur L. Samuel made this ideology a reality for the first time with his 1967 publication on a checkers game that learned by holding records of previous game states(Samuel 1967). This was a major development in machine learning overall as it used graph theory, heuristics, and generalization learning procedures, to help the system perform more realistically and have limited understanding of the moves based on optimizing rewards. With the rise in technological power, the idea of computer intelligence was substantiated with gathering data points for which computers could learn, like humans do. A notable and pioneering result of this was the inception of the image recognition database,

ImageNet, by Professor and computer scientist Fei-Fei Li and others ImageNet 2025b. ImageNet data was sourced from web browsers and was hand identified by humans into various categories. The training subset includes roughly 1.3 million images in total ImageNet 2025a. The ImageNet data set was pioneered to fix the gap within building "intelligent" models but not having the data, especially real world data to train them. It is also used as a benchmark to evaluate new and improved computer vision algorithms by supplementing and receiving resultant data recognition similar to humans. ImageNet was pioneered based off of psychologist and psychology Professor George Armitage Miller's 2005 pioneering WordNet dataset, a collection of English lexicographical pairings and connections between such words and phrases Miller et al. 1993. In its initial publication, WordNet had about 96,000 words and word derivations Princeton University 2025. This dataset was used to train Natural Language processors by fixing the issue of not having enough data to accurately implement John McCarthy's idea of trained models. With the influx of people's online presence and online data storage, more data could be collected more easily, leading to more machine training.

This trend of building bigger, and better datasets consumed the early 2000's. The idea with growing datasets and competitions for the largest dataset stem from how machines learn. It is easier to identify items through large sets of examples. When given an unknown problem, it must rely on the data to categorize, predict, or furthermore assess and perform its task. However, just like humans, models can manipulate data it has seen or encountered before. Unlike models, humans do have the ability to extract and reason logically even if they have not encountered a specified problem or fact before. This idea of a machine that can reason is outside the scope of this project's intentions and functionality. Taking this as fact, large datasets must be curated to cover these vast connections between objects, people, things and ideas so that machines can learn more complex "understandings" and respond correctly.

2.1.2 Current

Several datasets have grown since then to tackle different problems. Kaggle.com and University of California Irvine's (UCI) Machine Learning Repository host datasets ranging in size and topic. These are open to the public to train and test their own models. Hosted datasets span one of the earliest classification set, Sir Ronald Fisher's 1936 Iris plant classification at 105 B UCI Machine Learning Repository 1988, to credit card fraud detection, contributed in 2018 by Université Libre de Bruxelles at about 151 MB Machine Learning Group, Université Libre de Bruxelles 2025. Another large database is CommonCrawl, with slightly over 3e+11 Common Crawl 2025a cumulative crawl size [1] and is updated monthly with more internet data. It has petabytes of data consistently updated since 2008 Common Crawl 2025b. The actual quantification of datasets in terms of variables or dataset size is slightly ambiguous due to the type of data that is stored. A dataset of images will hold more storage than that of simple categorical or numerical data. However, the gross amount of data needed to fill such datasets can still be loosely understood based on the size and type of data collected. This is the metric that has helped the early dataset competitions, and continues to drive big data today. Even across these databases, there are few datasets based on court cases, especially state, civil, and pro se cases.

Collecting data on intangible items such as emotions makes datasets difficult to curate. In these instances, datasets steer away from trying to plot all the intricacies of a topic, and focus solely on quantifiable and identifiable attributes. In this way, legal datasets generally provide the most amount of data as possible on all quantifiable topics. A few forums such as the Caselaw Access Project (1.57 GB) and the Patent Litigation Docket Reports Dataset (5.6 GB) exemplify legal datasets and databases. These datasets cover court filings provided by school legal libraries, reporter notes, copyrights, patent information and much more. Seeing a need, the Caselaw dataset was initiated in an effort to make court data more accessible to researchers and everyday people. Their data is divided by various inputs via case, names, and other specifiers,

[1] A crawl is a process for having computer systems navigate through a wide range of the internet to look for information. A crawl size indicates all of the URLs, content, domains, host sites, and total pages analyzed from the crawl (Common Crawl Foundation n.d.).

thus making it not fully complete datasets, but fragments of such. Both datasets were made publicly available in 2018 and 2016 respectively, showing growing interest in the intersectionality of law, computation, and document preservation. Works such as "How Does NLP Benefit Legal System: A Summary of Legal Artificial Intelligence" published in 2020 (Haoxi Zhong, Chaojun Xiao, Cunchao Tu, Tianyang Zhang, Zhiyuan Liu, Maosong Sun 2020), and "Cross-Jurisdictional Data Privacy Compliance in the U.S.: Developing a New Model for Managing AI Data Across State and Federal Laws" published in 2024 (Okesiji et al. 2025), show the growth in interest and complexity of this interdisciplinary work. Patent research has been a common topic, as this information is widely available for patent filing purposes and searching, making it easier to conduct research on. However, the amount of datasets widely available to analyze civil court cases, cases within Georgia, and cases on pro se filers gets sequentially smaller. Other datasets and databases like the National Archive of Criminal Justice Data with the Inter-university Consortium for Political and Social Research host datasets on court cases and criminal activity, however still miss the pointed nature of the present project.

Cambridge University and Uppsala University curated a similar data set containing cases and tags based on court cases in the United Kingdom called the Cambridge Law Corpus. This corpus currently contains more than a quarter of a million cases, spanning from the 16th century to present day (Östling et al. 2024). They are individually tagged, with some being annotated by legal professionals for clarity. These tags included the following: case outcome, assigned court, year case was taken to court, judge, claimant, defendant, date of judgment, unique ID, case metadata, neutral case citations, and more as XML files and in a dataset. With the onslaught of machine learning and artificial intelligence, plus the general loose to non-existent formatting of cases in the region, this dataset serves as a conscious aim to provide many points of quality case details to train tools on. Whereas this corpus is mainly directory links to the cases, the aim of this project is to take it one step further and provide a fully polished dataset that is readily used for machine learning (Östling et al. 2024). Due to privacy concerns as this data was

collected directly from the government, only persons within London, and those who are doing research may access the contents of the dataset.

Internationally as well, Hanjo Hamann wrote and created a dataset for Germany's court cases from 1950 to 2019 in the paper "The German Federal Courts Dataset 1950-2019: From Paper Archives to Linked Open Data" (Hamann 2019). It holds a similar structure as the Cambridge Law Corpus but is more robust and open sourced via the custom project page. The page has since been updated with files including those in 2020. Unlike the Cambridge Law Corpus, this dataset is fragmented based on year and court type. It is also hosted across wikiData, with each wiki page corresponding to an individual item called "QID"s. This project received its data via multiple channels including the courts themselves and gazette archives. Although not a succinct dataset like the Cambridge Law Corpus and many other datasets, it does provide quick accessibility for online users, and research accessibility by providing its own JSON returning API. The project aims to illuminate and share the trends in judges decision making within common law and civil law, as it is an area not researched due to personalized ambiguity. Centered around the dispersed legal system of Germany, this dataset was conceived and breaks the connotations of a typical dataset. It can be seen that this trend of limited data due to legal intersection is not only seen in the United States, but also abroad.

2.2 Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation was originally instantiated by Patrick Lewis, et al. in 2021 in an attempt to update Large Language Models dynamically. They found that their frameworks outperformed other model types like parametric seq2seq, and architectures such as task-specific retrieve-and-extract, for which LLM's were previously used to build upon (Lewis et al. 2020). This process is conducted in three main parts: encoding, chunking, and scoring. For encoding, text is converted into numerical values based on various characteristics and the model type. These encodings are then broken up into chunks or sub sections based on model maximum intake or a user set maximum. These chunks of values are then scored to generate a parametric

[2] response. This framework has gained traction, with sufficient research being done on each step of the process to optimize models. Chi-Min Chan et al. wrote in 2024 the process of refining the queries itself to develop better systems (Chan et al. 2024). Similar to the original problem presented by Lewis and others, models can only understand what they are given and not what is interpreted. This extends to the queries given to a RAG system, as variations of the same question can produce different semantic encodings, and thus be matched to different responses. Since queries are also encoded with the same encoder for the main text, it is also subject to the same issues. Also, the queries themselves can indicate dense interconnected dependencies that need to be processed and understood, rather than formulaically calculated and cross checked with a text. Their process involved developing responses at each step of the RAG process, so that the model can dynamically try to deconstruct the queries given state based information. Their system included three query processes: rewriting the query, decomposing the query into multiple questions, generating response directly, or disambiguating the query, and has been proven to out perform other question and answering task frameworks (Chan et al. 2024).

Given the properly formatted question, RAG systems must also encode the document or text of choice. In doing so, frameworks can only encode a given amount of text at a time. For short texts, there is not much context to decipher, making a standard RAG framework acceptable. In cases where there are longer texts, or texts that are even slightly over the bounds of the methodology, the text must be chunked. To be chunked, texts are broken down into more manageable sub-pieces of the main text. Long documents pose issues as the length of the passages is inversely related to the quality of generated output after a certain point, making them less useful when applied directly in other pipelines (Jin et al. 2024). In 2025, Cesar Abraham Gomez-Cabello and others identified ways in which the text chunkung can also affect the RAG process that is part of the base of Jin et al's problem (Gomez-Cabello et al. 2025). Gomez-Cabello and others examined various chunking styles such as length based, semantic based[3], and text structure

[2]Parametric computation was developed by Luigi Moretti in "Architettura Parametrica" in which computation was given a set of rules and relationships to be able to map or reduce larger computation and inputs based on the rules given. (Canestrino 2025)(Idea from Moretti, 1940-1942)

[3]Grouping sentences and words based on how closely related they are to one another.

based chunking. Each method resulted in slight variations of RAG responses due to the fact that the model was understanding the texts in different ways. In the areas of precision, recall, and F1 scores, adaptive chunking scored the highest, holding respective values of 0.5, 0.875, and 0.643 in each category (Gomez-Cabello et al. 2025). In this instance, recall measures how many documents were retrieved to answer the query, while precision measures how many of those documents were actually relevant to the given query (Agnihotram and Sarkar 2025). F1 corresponds to a calculated combination of these values that assess the model in both contexts. The reported values correspond to 0.1 or more score difference between adaptive chunking and the next highest scored chunking methods. Their work showed that adaptive chunking increased the performance of RAG-LLM outputs compared to the other chunking strategies. This work is corroborated by Urho Lensu in their 2025 piece "Impact of chunking Granularity on Accuracy and Token Consumption in Retrieval-Augmented Generation for Question-Answering" (Lensu 2025). Although using a sliding window approach, one in which text is broken up into overlapping segments, this can even affect the results of RAG by changing specific token sizes and strides. Across the two RAG systems built for the study, moderately sized chunks performed the best when their output was semantically compared to the expected output (Lensu 2025). This in itself has drawbacks on memory, storage, and computation as Lensu found slight increases in ROUGE accuracy from a 15% to 115% increase in token consumption. As a highly sensitive model yet powerful tool, the factors that go into each step of Retrieval Augmented Generation requires intentionality that affects this young method as a whole.

RAG has specifically been developed to assist legal practices as well, similar to the objectives of this project. Rahman S. M. Wahidur and others introduced Legal Query RAG or LQ-RAG that is feedback driven and aims to mitigate complexities within legal document parsing (Wahidur et al. 2025). They achieved this by adding an evaluation agent that analyzes responses within the model, a specialized text generator, a prompt engineering agent, and a LLM finetuned for legal embeddings. Tasks were split between a fine tuning layer which adjusts the embedding LLM and generative LLM. The RAG layer computes similar to a standard RAG system, with

advanced mechanics and agents to retrieve optimal results. It is within this last layer that a proper response is generated and given. This process also involves calculations using Multiple Negatives Ranking Loss (MNRL) in forward passes to increase the similarity of chunks to other similar text inputs, while also enlarging the gap between dissimilar inputs and texts (Wahidur et al. 2025). Also different for other RAG models, LQ-RAG updates the weights to include the set of parameters with a size smaller than that of the original weight calculation. This allows for more specialized knowledge for that particular domain. LQ-RAG uses other methods such as Facebook AI Similarity Search, ChatGLM-6B LoRA 16-bit instruction fine-tuning, OpenAI GPT-4, Best Matching 25, and other methods to develop this type of RAG (Wahidur et al. 2025). Their work highlights the interdisciplinary application of RAG while also noting the flexibility of it to be combined with specialized LLMs and other frameworks to achieve optimal results. It also highlights the quick pace that RAG is being retrofitted into the computer science field, as this paper and methodology was published 4 years after the inception of RAG by Patrick Lewis and others.

2.3 BERTScore

RAG is a new methodology in a history of other sectors of Natural Language Processing, machine translation, and artificial intelligence. Tests such as BLEU (Bilingual Evaluation Understudy) (Papineni et al. 2002), METEOR (Metric for Evaluation of Translation with Explicit Ordering) (Banerjee and Lavie 2005), and ROUGE (Recall-Oriented Understudy for Gisting Evaluation) (Lin 2004) have been used in testing the accuracy of text generation produced by the aforementioned fields. Tianyi Zhang, et al. developed a new methodology in 2019[4] specifically for comparing BERT text embeddings using cosine similarities of token embeddings between two sentences (Zhang et al. 2019). The sentences comprise of the target sentence and the output sentence from the model. Their findings showed that many iterations of BERTScoring outperformed most if not all categories of testing across various language translations, training methods, Pearson's Correlation coefficient testing, and other tests (Zhang et al. 2019). A close

[4]It has since had updates to the paper again in 2019 and once more in 2020.

competitor was the RUSE methodology, however, is not as widely available or accurate as the BERTScore. However close, BERTScore outperforms matching exact, paraphrased, or similar text stems by utilizing pre-normalized vectors, greedy token by token matching similarity checks, and pairwise cosine similarity between embeddings (Zhang et al. 2019). It represents the similarity via three outputs; the recall (R), precision (P), and F1 scores. The R score matches cosine similarity from the target to the output sentence, while P computes cosine similarity from the output to the target. F1 is a combination of the recall and precision values, and thus an accurate measure of how the model runs similarly in both directions (Zhang et al. 2019). The formulas from their paper for each output are shown below.

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^\top \hat{x}_j$$
$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} x_i^\top \hat{x}_j$$
$$F_{\text{BERT}} = 2 \cdot \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

The variables are as followed: x is the target text as tokens, $\hat{x}$ is the output text as tokens, R_{BERT} is the R score, P_{BERT} is the P score, and F_{BERT} is the F1 score. Comparable to other methods, rare words specifically pose challenges to this type of problem. BERTScoring addresses this by having the possibility to add weights to words, thereby increasing similarity contextually (Zhang et al. 2019). Any new metric or methodology comes with analysis, proofs, and testing to ensure it is a valid method and to identify any pitfalls. Michael Hanna and Ondřej Bojar analyzed the BERTScore within the next year of the method being published. They identified grammatical and linguistic word ordering that can vary the results of a BERTScore, which is similar to the findings of RAG system inquiry inputs (Hanna and Bojar 2021). Sentences with grammatical or lexicographical errors did not compute as well, even if the base of the sentence is sensible. Despite these differences, it does state many instances where BERTScoring outperforms other methods holistically (Hanna and Bojar 2021). However in cases where the output text was stylistically similar but has a high lexical overlap to the reference text, the

BERTScore struggled in scoring the texts (Hanna and Bojar 2021). This led to the conclusion that BERTScore is better at scoring texts that are stylistically different from one another (Hanna and Bojar 2021). Although true, the BERTScore did accurately score their grammatically error prone dataset in all conditions set out within the paper. As a young methodology, these pitfalls are to be expected. The BERTScore does show high promise as even within these, it outperforms other metrics.

2.4 Dataset creation

2.4.1 Ethics of Law and Machine Learning

A caveat to making datasets is understanding, and respecting the object being observed. Producing a dataset of civil court cases is a direct insight to people's everyday lives and decisions. By creating this, care must be taken, similar to the Cambridge Law Corpus (Andreas Östling, Holli Sargeant, Huiyuan Xie, Ludwig Bull, Alexander Terenin, Leif Jonsson, Måns Magnusson, Felix Steffek 2023). Their dataset of similar cases is only accessible to specific types of people, and those only within London, to protect the privacy of people's cases mentioned in the dataset. It is the responsibility of the dataset creator to identify reasons for collecting, processing, and reproducing such figures, and to ensure their data meets basic guidelines. Structures exist across some state legislatures on data collecting, manipulation, and sharing. Using the California Consumer Privacy Act (CCPA) in California as an example, people have the right to know what data of theirs is being used, to opt-out of data inclusion, correct, limit usage of, and expunge their own data (*California Consumer Privacy Act of 2018* 2018). These laws mainly pertain to current collection of data that is done through user-company interactions, but does not explicitly scope to publicly available sources. The source regulation may be due to the nature of the item which data is collected, like home size on Zillow, or public image. Responsibility still relies on the curator of how this public data is utilized. SerpApi, a third party API for Google Suite products, explicitly states, "SerpApi assumes liability for the lawful collection of public search

data (scraping, parsing, and related actions), but not for how that data is ultimately used," (SerpApi, LLC 2026) keeping in line with the responsibility of the curator.

A secondary ethical dilemma is with scraping websites with explicit robots.txt files. These files outline the allowed and disallowed usage of scrapers to parse a specific website. Along with the terms of service, many methods can be blocked and or legally challenged if violated. An example of this is Google Scholar being a "Disallowed" site to be scraped. All other Google Suite applications and functionalities within the company can be found https://www.google.com/robots.txt. Using the requests package in Python, it would be simple to spoof the address of requests by changing the header and specifying diverse user-agents [5]. Doing so bypasses the identification that requesting multiple searches are coming from the same automatic script, and looks more human processed. Although it can be done, one must respect the robots.txt in place, and not leverage the potential motives of collection for the explicitly stated rules in place.

On a larger scale, exceeding server limits can be seen as invoking "unauthorized access" of the website. On top of site access, exceeding such usage poses a problem federally. In conjunction with the previous articles of ethics presented, the full cycle of data collection, wrangling, and usage, must legally comply with the company and website rules, state laws, federal laws, and the data producer simultaneously. Scrapers have legally challenged these ideas and helped form the precedents used to this day. HIQ LABS, INC.,Vs LINKEDIN CORPORATION. (United States Court of Appeals for the Ninth Circuit 2022) defines that scrapers are not in violation of CFAA given that the site is public (Berris 2023). This presents a basis of accessibility for scrapers generally, before looking specifically at a site's robots.txt. Enacted in 1986 on top of a previous cyber crimes law, the Computer Fraud and Abuse Act (CFAA) (United States Department of Justice 1986) was put in place to legally and ethically define provisions of internet crimes, including web scraping and hacking (National Association of Criminal Defense Lawyers n.d.). Via CFAA, 18 U.S.C. § 1030., exceeding server limits can be seen as invoking "unauthorized access" of the website and can be classified as fraud or abuse (United States Department of

[5] A user-agent identifies the machine trying to access information or visiting any site (MDN Web Docs Contributors 2025). This can be changed within the header to spoof where the request is coming from.

Justice 1986). Violations such as this can occur automatically if information is derived from a non-public computer or accessing a site above the granted limits. In conjunction with the previous articles of ethics presented, the full cycle of data collection, wrangling, and usage, must legally comply with the company and website rules, state laws, federal laws, and the data producer simultaneously.

Despite these ethical dilemmas, datasets of court cases are still possible to produce. Major issues with legal datasets involve the lack of motivation, and heavy privacy concerns and protections around legal cases, including but not limited to sealed records and closed court proceedings. This in itself limits the amount and type of data that can be compiled into a dataset while also respecting the cases and court system. Under this guise, it is still possible and evident that many prior works trained small models on similarly small legal datasets that include private data without proper licensing usage (Zheng et al. 2021). This project takes the next step by being among the first to recognize, organize, and curate a large open-source, legal-focused pre-training dataset (Zheng et al. 2021). To combat this issue of limited scope and size of legal datasets particularly, new efforts that will not be discussed have arisen to solve this problem.

Due to the various legal protections already in place, the United States was chosen as the base for case content. In the United States, federal case information and documents are generally freely made available to the public. State cases are harder to find publicly, as they are normally smaller cases and to protect the privacy of everyday people. Searches for either type of case comes with the caveat of knowing some case detail, whether it is case name and or case number with other identifying markers, or paying to view other information.

2.4.2 Algorithmic and Dataset Bias

The training data, algorithmic restraints, and physical limitations are the main reasons machine learning algorithms fail or struggle to make complex decisions like humans. Algorithms consume bias, tendencies, and imperfections from training data due to the fact that algorithms only have the context of the dataset to learn. In this way, any biases and issues within a dataset will be

calculated by those same trends and reflect results with those biases. Forms of bias include toxic phrasing in legal documents which lends itself to practitioner bias Henderson et al. 2022. The Georgia Special A.I. Committee agrees that many machine learning models echo the biases that exist within the training data Supreme Court of Georgia 2025b.

Another form of such bias is sampling bias, in which the sampled group does not encompass the depth nor range of possibilities in the target population. This can be seen in issues with the current healthcare system and emerging healthcare technologies, with tests for diseases only coming from a particular subset but used to model the whole population Nazer et al. 2023. This project will slightly reflect these biases by having a small pool of sources and reflects the ability for even their resources' ability to gather data. A mitigating factor would be to directly receive case data from the courts, as there would be no limiting factor in the cases they provide.

Data can also suffer from labeling bias, in which data is misrepresented because of irregular testing, limited resources, or misunderstanding of issues faced by subsets. Inter-correlated events and attributes contribute to the results in various activities and people. If such events are not taken into account, like lack of Canadian race and disability data due to no standardized identifiers for it (Pottier et al. 2024), accurate analysis can not be conducted. If it is conducted, it misses key features and complexities seen in real life. Dataset curators must be keenly aware of these facts when developing and reporting their work so that it limits inclusion of these biases. This project does present a gross instance of labeling bias as it does not collect nor assess the complexities of a court case, or label cases in what stage of the judicial process it was in. Due to the complexity and timeline of this proposal, this bias was not actively mitigated and should be considered within future work.

2.5 Why Georgia?

With the cases in New York and Texas listed earlier in which lawyers have been fined for usage of AI in producing fictitious citations and the general spike in such technology, Georgia has looked into the relevance of such technologies within its own courts. Georgia established a

judicial committee in August 2024, called the Judicial Council of Georgia Ad Hoc Committee on Artificial Intelligence and the Courts, specifically to review the place and ethics of new generative artificial intelligence within the court system Supreme Court of Georgia 2025a. They published a full report in July 2025 identifying the findings of their researchSupreme Court of Georgia 2025a. In the results, it suggests having dedicated AI resources that are pre-approved and acceptable for use within the courts, and to have more technical research done by professions on various cases (Judicial Council of Georgia Ad Hoc Committee on Artificial Intelligence and the Courts 2025).More results are listed but all point to developing some adaptive form of the current court system that upholds the integrity, rules, and privacy of the court rooms. Georgia's Bar committee has also pledged to develop a special committee established in October of the same year, called the Special Committee on Technology, Artificial Intelligence, Tools, Resources, and Legal Obligations (State Bar of Georgia n.d.), to identify the same issues. They have come up with regulations that only allow machine learning to be used if the code to build such models is also submitted to the courts for the case it was used in Court of Appeals of Georgia 2025. This faces a problem with using large models like ChatGPT, Google Gemini, and CaseLaw which have privatized code bases.

Generally, Georgia is experiencing a rise in cases of all kinds across the state. In 2024 and 2023, Georgia was the 5th highest caseload per capita by state National Center for State Courts 2025. It is preceded by Texas, New Jersey, California, and Florida. Georgia has slowly grown each year on the list, reaching closer to the highest caseload per capita. It is important to examine this growth and curate the proper technologies to adapt to this. It is also the case that various states hold various court structures and laws regarding the same topics. By restricting the cases to only one state enhances the predictive and analytical abilities at a higher level due to the lesser amount of variables to factor. With higher case volume, emerging research into technological advancements, and geographical consistency, Georgia was selected as the state of choice.

3
Methodology

3.0.1 General Overview

Each script was exported as its own Comma Separated Values (CSV) of cases developed from the API, RAG, or web scraping. This allows for future source-to-data output analysis where warranted. The datasets were joined together under the matching variable tags using an R script. Missing data was filled with a "." value. This was necessary in the event data was not applicable or found for that specific case. Using a period instead of nan, NaN, NULL, or any other value would convolute methods of comparison due to varying types and package capabilities. The string of "." can be evaluated as text and assessed accordingly. The GitHub link to all code and output can be found here: https://github.com/110nyLaw/GaProSe

3.1 Identify Data Sources

Court cases and their decisions are typically hidden behind paywalls or authentication processes. Although free, many sites also require one or more specific pieces of case specific information, such as case ID, to be known for any search to occur. Sites that did require a pay wall or only had case specific information search were not included in this set. To keep in line with ethics, privacy for involved parties, and inaccessibility to such cases, these cases were not included in the dataset. As long as the source sites were free, had minimal limiting factors, and hosted some

Georgia civil pro se cases, it was examined for the development of the dataset. Four sources, which included JudyRecords.com, https://static.case.law/ , www.courtlistener.com, and Google Scholar Case Law search engine via SerpApi, fell into these criteria and could be scraped. These sites were found from Google searches and following recommendations to databases within each website's structure. These sources all provided data in varying formats for which this project aimed to extract, clean, and recombine.

3.2 Tag Development

Developing which tags to include was dependent upon case type, identifying markers, previous research done with legal datasets[1]. Some of the tags developed included basic court descriptors which included case name, case number, judicial officer, court, and case type. Additional tags included were dependent on the information provided on the services, case identification markers, plus other precedent in this kind of research (Andreas Östling, Holli Sargeant, Huiyuan Xie, Ludwig Bull, Alexander Terenin, Leif Jonsson, Måns Magnusson, Felix Steffek 2023). Identifying markers such as the case name, judicial officer, and case number were identified for each case. Other tags were included but are only applicable for specific case types. These include compensation, serve date, and petitioner. The following 16 tags included are as follows: Case name, case number, Petitioner, Defendant, Judicial Officer, Date filed, Case type, Case status, plaintiff, Compensation, Court, Serve date, Compensation, Chronology, and data location(s).

To preserve the ethics of the cases and the privacy of the involved individuals, identifying markers such as addresses of any party were not included. Access to cases with redacted or changed names typically are sealed and not released to the public, and were not accessible for this study. Cases developed in this study did not reach a level of sensitivity to require such actions by the court, and were publicly available, allowing them to be ethically used within this study. Overall, information on plaintiffs and defendants can prove insightful for understanding the relationships between groups and the power dynamics that can affect their case.

[1]See (Locke and Zuccon 2018), (Leitner, Rehm, and Moreno-Schneider 2020), (Xiao et al. 2018) and others mentioned within the text for relevant research

3.3 Web Scraping with Selenium and Beautiful Soup

Based on the source, different ways of extracting data were necessary. Some of the webpages used are themselves databases, which hide some results due to the cost of upending all of their case data. Due to this, some of the websites have API's in which one can retrieve larger sums of raw data from their collection. Google Scholar , CourtListener, and JudyRecords.com have APIs to use. JudyRecords and CourtListener have custom APIs, while access to Google Scholar can be accessed via SerpApi, a third party Google Suite API. JudyRecords.com didn't respond to requests for an API key. This led to generating a custom python script to scrape the site. Scripts were developed under the following tools: used Spyder IDE, packages [os, time, transformers (AutoTokenizer and AutoModelForQuestionAnswering), pandas, torch, torchmetrics.text.bert (BERTScore), random, selenium (webdriver and webdriver.common), bs4 (BeautifulSoup) and requests in versions as of March 30, 2026.

The first website was JudyRecords.com, which hosts and grows a collection of court cases across the United States. These cases are accompanied by basic information such as case number, court type, plaintiffs, and a timeline of documents filed thus far. This was used as the base for the dataset as it provides a plethora of almost full information on each case. This website has an API, however, efforts to obtain an API key were left without a response from the contact listed. The lack of API forced the usage of Python methods to extract necessary information from the HTML. Without a robots.txt file indicating inaccessible operations, basic safeguards such as rate limits were put in place.

JudyRecords was first viewed through Safari webpages to access Apple's Developers "Show Page Source" command. This was examined and taken into account to identify structure, information extraction, and query structure. Upon having identified structure and predetermined variables, Selenium was used to dynamically generate the base page and input the search queries. JudyRecords, like search engines and many other websites, dynamically loads data upon requests, which called for the use of Selenium. Search parameters were developed based on the website search criteria and format, and expanded to get a larger scope of Georgia's Civil

Pro Se cases. JudyRecords has a tab for site specific search tips in the requested formatting. The following 11 search parameters were used to search the sites: “georgia court civil pro se”, ”georgia civil pro se”, ”georgia civil pro se -statement of claim”, ’divorce pro se ”georgia court”’, ’incarcerated georgia civil court ”pro se”’, ”’pro per” ”georgia court”’, ”georgia court civil pro per”, ”georgia civil pro per”, ”georgia civil pro per -statement of claim”, ’divorce pro se ”georgia court”’, and ’incarcerated georgia civil court ”pro per”’ in the given style. The various formats include information about specific types of civil cases or the wording of such that were not produced within the first few searches. Once the search was dynamically loaded through Selenium, that page was dissected to find the hypertext reference, hereby mentioned as href. Href’s held the necessary information for tags which was further extracted with BeautifulSoup. Each href was followed and necessary data from the destination was extracted. Information was immediately stored into a Python pandas data frame which would be exported later. Selenium was used again to paginate through all the results of the initial query, and to move onto the next query until all search parameters and pages were exhausted.

3.4 API Usage

API calls were used to develop the rest of the dataset through CourtListener and SerpApi, which hosts API endpoints for Google engines. These websites supplemented the data set by adding more cases not initially under JudyRecords. An API key was obtained to request data points from each of the website’s databases. Documentation on each page identifies specific parameters and API setup for successful extraction. Preferences were set to receive JSON objects from the API calls for cleaner data extraction. The queries requested from CourtListener were similar to those of JudyRecords. They included the following queries: [”civil pro se”, ”civil pro se statement of claim”, ’divorce pro se’, ’incarcerated civil court ”pro se”’, ’civil pro. per.’, ”civil pro. per. statement of claim”, ’divorce pro. per.’, ’incarcerated civil court ”pro. Per.”’], and the following courts within the parameters heading: [”ga”, ”gand”, ”gamd”, ”gasd”]. Unlike JudyRecords.com, CourtListener also has cases organized by court which is helpful for API

content specificity. These queries received the requested data and their value was stored in a pandas dataframe. These were put into separate scripts, as each would pull data independently and stored in separate CSV's.

For SerpApi and Google Scholar specifically, the parameters only included "georgia civil pro se" and an internal state identifier of "4,11". The "4" represents a required value and "11" represents all Georgia court cases hosted on the site, whether they are appeals or Supreme Court cases. To remain in line with Google Scholar's robots.txt, the links were not searched with Selenium. This resulted in the dataset to only note web addresses and the returned JSON object. The links to the GoogleScholar case were included in the JSON object.

3.5 RAG Development

3.5.1 RAG Document Retrieval

Before implementation of the RAG model, it must take in data. This data will be full legal documents provided from a subset of Harvard Law School's Case Law Access Project. Their platform hosts a section dedicated to downloading batches of data stored on their website. It includes case documents as local HTMLs, corresponding case JSONs, and metadata for the folder as a whole. The specific folder downloaded and used to test the model was volume 1 from Savannah, Georgia. Using BeautifulSoup and file operations, the HTML folders were opened and all text was extracted into a dictionary for each document. Data was imported with the file name as the key and content as the value. The dictionary was then restricted to be exported as a CSV to reduce unnecessary compilation and for direct usage to the RAG system.

3.5.2 Model

In a separate file, load the CSV from reading the HTML files and initialize the tokenizer and auto model for question and answering with "deepset/bert-base-cased-squad2" (deepset 2024). The model is a BERT base cased[2] model trained on SQuAD version 2 and specifically utilized

[2]Bert models are case sensitive. Uncased models convert all text to lowercase before computation. Cased models leave text in upper and lower case.

for extractive question and answering. BERTScore is resource heavy and takes an extended time to train, which is why gradients were turned off. A set of questions were then initialized that are presented in the dataset, similar to the set up of the other datasets created. The questions are as follows: ["What is the case number?", "What is the case name?", "Which court is the case filed under?", "Who is the judicial officer of the case?", "When was the case filed?", "What is the case type?", "What is the status of the case?", "Who is the plaintiff?", "Who is the petitioner?", "Who is the defendant?", "How much was compensation?", "When was a party served?", "Who is the pro se filer?"]. Chronology and citations were not included in this rendition due to the complexity of returning multiple answers for a given question. They require a level of information and chunk scoring that is either unavailable or not currently implemented. For each case of local HTML cases and each question in the dataset, the question and case were tokenized.

Once the question and context were reduced to tokens, the context was then chunked to find the best response. This model uses a sliding window chunking approach due to simplicity of programming. This method chunks tokens of the text, with some overlapping stride with other chunks. The stride sets up the overlap and the max length sets up the maximum length for which chunks will be stored. For implementation, only the second input (case content) was truncated and padded to max length in case there aren't enough tokens to fulfil requirements. Also, max length was set to 384, stride to 128, and overflowing tokens were returned as "pt" or pytorch tensors to enable chunking. Although the model can support chunks as large as a maximum length of 512, it was seen that a smaller token size increased correct output. The last layer of the model without gradients was extracted and used to compute the beginning and ending logits, or raw prediction values, which was held in variables called startScores and endScores. These helped locate the chunks with the corresponding answer, according to the model. Given the fact that the input context is normally larger than the 384 tokens that max_length was set to previously, tokenization returns multiple tensors corresponding to truncated chunks of the original document. Given the start and ending scores, the index of the chunk that has the

highest similarity was stored. Since the response can be split up across multiple chunks, the start and end indexes are checked and swapped if the start was higher then the end index[3]. These indexes were used to locate the best section of tokens within the chunk from "input_ids". The tokens were then decoded to receive the model's best answer to the given query and is appended to the other answers for the given document. These processes were repeated for each question and the whole process for each document read in.

3.5.3 BERTScore Analysis

To get an accurate comparison of the RAG system, the same cases fed into the model were hand verified. This entailed going through each document and extracting the correct, or perceived correct[4], output for each variable question. This verification was extracted as a CSV and read in alongside the RAG CSV output. General cleaning operations like ordering, renaming, and such occurred. Due to the setup of previous sections, the CSV's have the exact same number of variables and positioning. A variable list of lists was also initialized that held the following structure: [["Variable", 0, 0, 0]...]. This structure was used later to index and calculate the mean precision, recall, and F1[5] of each variable.

Since the model used is not largely supported, the number of layers and the language were defined. Within the model definition, idf was also set to True, to weigh common input more harshly. This essentially normalized the model to not account for punctuation and not enough tokens for comparison. Each row was flattened and converted to lists so the model can recognize text input instead of as a dataframe row. After converting to a list, the model is diverse enough to take in the full rows of both hand verified and RAG output as the reference and candidate contexts. Semantic comparisons were run across both lists, returning BERTScores as a tensor of the same amount of items. Scores were grouped together by variable and stored as (Precision,

[3]Occurs frequently with sliding window approach.

[4]Human verification of cases is subject to scrutiny as humans can make mistakes. Perceiving a correct encompasses the efforts made to verify items that are incorrect.

[5]The R (recall) score matches cosine similarity from the target to the output sentence, while P (precision) computes cosine similarity from the output to the target. F1 is a combination of the recall and precision values, and thus an accurate measure of how the model runs similarly in both directions.

Recall, F1) tuple representations, with an extra column noted for the means of each metric. By doing so, a holistic view of the case scoring and the per variable can be further visualized. Full columns with a case name as the key were added to a dictionary which will transform into the necessary dataset.

While analyzing each variable, the corresponding scores were added into the variables list of lists that hold the question based P, R, and F1 values. To get the score of how each variable performed, the resultant value of P, R, and F1 and were divided by the amount of cases and was added to an extra column called 'avg'. The index of the BERTScore and the variable list of lists are identical, leading to the means to also be identically placed within the dictionary. To get the score of the overall model, summed means of P, R, and F1 just found were divided by the amount of variables present. As the final element, the overall mean score just computed was appended. This concluded the analysis generation. The resultant exported CSV contains the case names along the left, variables across the top, case BERTScore means down the right, and variable means across the bottom.

4
Results

The total dataset came out to be 10,704 cases cases and all spanning from 1866 to 2021. 1,036 cases came from Google Scholar/SerpApi, 315 from CourtListener.com, and 9,353 from JudyRecords.com with values over 15 variables. This accumulated to 10.2 MB of data, only with JudyRecords and CourtListener. Google Scholar only had 3 variables (case name, url, and JSON) due to accessibility issues which is why it was left as a separate dataset. Out of the total 154,688 data points in the combined dataset, 9983 data points or 6.45% of the dataset is unavailable or missing. A composite of all the BERTScores was generated into an extractable CSV format which can be found in the gitHub Link mentioned in the Methods section. The following is the average precision, recall, and F1 score for each question type can be found in Table 4.0.1.

The resultant precision value was 0.590, recall was 0.557, and F1 was 0.556 across the whole model. This gives the model an overall F1 accuracy of 55.6%, meaning that the cosine similarity of the target text to the output text was more tangentially similar across the whole model. Document by document findings can be found in the file output of ragEval located in the gitHub link in the Methods section.

To test the scalability of the project en masse, all documents were run on two machines, one a Macbook M2 chip Sequoia 15.3.1 kernel Darwin 24.3.0 and one an Ubuntu 24.04.4 LTS

(GNU/Linux 6.17.0-19-generic x86_64). The Ubuntu machine was run on a virtual environment through the terminal while the Apple machine was run in Spyder directly. The run time for each file is seen in Table 4.0.2.

The run time for the SerpAPI file is not fully indicative of the time taken for the local machine and the API to communicate and run each step. After each paginated search, a delay of 1 to 30 seconds is implemented so as to not overload the host. With about 50 pages requested and searched, that inflates the time found by at most 900 seconds (15 minutes) or at a minimum 50 seconds ($0.8\overline{3}$ minutes).

Tests were also run that show the differences in queries and the output generated. The original questions and RAG output can be seen in Figure 4.0.1. he response only found three answers: 'trespass', 'Very', and 'Thomas Heery'. These questions were utilized to develop this program. After questions were slightly altered, the RAG system did not find any matches within the text to answer the questions, as seen in Figure 4.0.2. This version of the same questions led to no text output from the document.

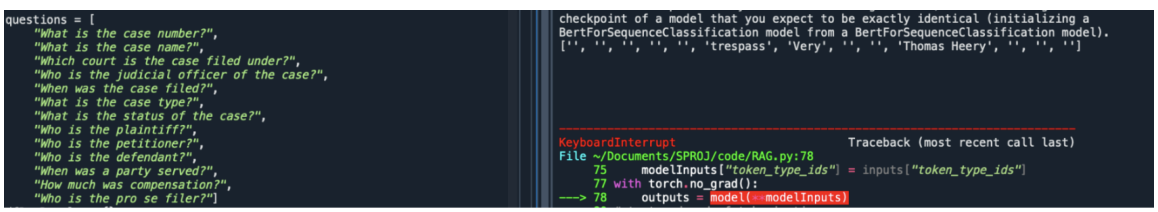

Figure 4.0.1: These are the questions and the subsequent output for 0001–01.html. The response only found three answers: 'trespass', 'Very', and 'Thomas Heery'.

Figure 4.0.2: These are the altered questions and the subsequent output for 0001–01.html. The response found no viable answers.

Variable	Average Precision	Average Recall	Average F1
Case number	0.655	0.655	0.655
Case Name	0.654	0.654	0.654
Court	0.654	0.654	0.654
Judicial officer	0.645	0.625	0.626
Date filed	0.629	0.613	0.619
Case type	0.619	0.611	0.612
Case status	0.589	0.560	0.565
Plaintiff	0.500	0.427	0.441
Petitioner	0.534	0.442	0.449
Defendant	0.461	0.359	0.367
Serve date	0.523	0.452	0.421
Compensation	0.553	0.541	0.518
Pro se filer	0.651	0.649	0.647
Average	0.590	0.557	0.556

Table 4.0.1: BERTScore Values Across all Variables

File Name	Mac	Linux
scrapeGoogleScholarLaw.py	1808.882 seconds (30.148 minutes)	1827.144 seconds (30.452 minutes)
scrapeHTMLFiles.py	0.131 seconds (0.002 minutes)	0.106 seconds (0.002 minutes)
linuxJudyRecords, macJudyRecords	4634.509 seconds (77.242 minutes, 1.287 hours)	3438.649 seconds (57.311 minutes)
scrapeListener.py	1651.568 seconds (27.526 minutes)	1722.740 seconds (28.712 minutes)
ragEval.py	51.536 seconds (0.859 minutes)	36.739 seconds (0.612 minutes)
RAG.py	1937.635 seconds (32.294 minutes)	1136.370 seconds (18.939 minutes)
Total	10084.261 seconds (168.071 minutes, 2.801 hours)	8161.7478 seconds (136.0291 minutes, 2.2672 hours)

Table 4.0.2: Execution Time of Project Documents over Two Different Operating Systems, Time given in seconds, minutes, and hours where necessary

5
Discussion

5.1 Interpretation of Results

5.1.1 Dataset

The dataset itself is small in relation to other datasets. At 10.2 MB and about 11,000 cases, it is a fraction of the results seen in CommonCrawl, the 5.6 GB Patent Litigation Docket Reports Dataset, and other legal datasets. The comparison of resources and funding allocated to this project versus other larger datasets of similar ideology also can help scale the contributions this project has committed to. Whereas this dataset is highly specific, both in geography and case type, the other datasets span many geographies. This alone expands the amount of data possible, and grows the size of the dataset itself. It can easily be scaled to different states and with varying queries to gather more state by state analyses. Given the run times of each query however, it can take large amounts of time to parse this expansion to more states. Scraping Judy Records did take up most of the time used to run all the files, accounting for about 45% of run time. As JudyRecords is responsible for about 87.4% of the data, scaling this scraper will add significantly more cases than expanding the API calls for CourtListener.

Although it may provide some information about each case, there are still some missing case details that would add depth of understanding. An example of this is the identification of possible class actions. The presence of multiple plaintiffs may be an indicator of this, or the

evidence of multiple cases against the same company that opted out of a class action. Some companies such as Wilson Young Rentals and the World Finance Corp of GA are mentioned several times as Defendants that have cases within the dataset that were filed within days of each other. This could potentially be the result of entities opting out of a class action or simply filing cases within the same general time frame. This question is not immediately understood within the dataset. More information would need to be added in order to understand this.

The data set would further need to be cleaned as some unnecessary input is also within some of the fields. An example of this is seeing "File Date" within the responses for "Date Filed". This would cause issues when simply looking for a date object and encountering a string instead. Also, this convention is not standardized across the whole dataset. This convention is particular to the JudyRecords cases and varies for the other inputs. It helps identify from first glance where a case was scraped from, but not helpful if applied in the current state for modeling.

Given the data that is present, having roughly 7% of the data missing is an acceptable measure. Many of the missing values seem to come from the "Citations" from JudyRecords cases which should be a list of all cases the given case cited. For many, the structure of the data with this information was not provided, and therefore not input into the dataset. The other missing values which included Plaintiff, Petitioner, Defendant, Compensation, Date Served, and Chronology are from CourtListener cases. Although the dataset does provide cases, it's immediate usage within machine learning is not advised until it is cleaned and some more missing data is filled in. More information about each case and their relevance to other cases would need to me documented and input into the dataset before being utilized.

5.1.2 RAG Model and BERTScore

The model does the best job at precision, meaning the RAG answers were mathematically similar compared to the human verified responses. Recall and F1 were relatively the same and slightly lower than the precision score. In conjunction with about a 0.56 F1 score, the model did not accurately provide responses to most of the variable queries. This discrepancy can be seen in

the RAG output of file 0383-01.html. When asked what the case name was, the response was "Legatt vs. Tollervey" while the human verified response was "Howland, Ward & Spring vs. John I. Dews, exâ€™or of W. C. Butler.". Looking back at the HTML, "Legatt vs. Tollervey" does not show up in the text at all and received 0.000 for precision, recall, and F1. On the other hand, 0298-01.html RAG output for case name was "." with a human verified response of "Commissioners of Pilotage vs. John Low and others.", and received a 1.0, 1.0, 1.0 for precision, recall, and F1. Between these two examples alone, there exist discrepancies among the RAG outputs, RAG document and chunk referencing, and the BERTScore which is showing varying similarities. Given that "." is not comparable to the case name and with idf set to true, receiving a high similarity for this query is misleading. These similarities, whether initially true or not, are what comprise of the overall P, R, and F1 scores even through they do not represent the similarities stated. On the other side, the original output for the case name of 0383-01.html poses the first set of problems. The output case name not being within the text shows issues of referencing as "Legatt vs. Tollervey" shows up in 0228-01.html instead. Given this mix up, the model continued to reference the incorrect texts, as "Great Falls Manufacturing Company vs. Mathers, 5 N. H. Rep. 574. (7 Amer. C. L. Rep. 114,)" was the output for case name in 0228-01.html and isn't within this text but within 0505-01.html. Although the model should only be referencing a single text at a time, it is reaching into other texts for possible answers. Both the improper output and semantic similarity are misleading metrics.

As a system, RAG is highly sensitive to the encoding types, inquiry structure, how data is read in, and chunking architecture. Although the model overall shows mediocre promise, individual questions need to be checked before being put into use elsewhere, if using the model as it currently stands. Reliance upon this model in the current state can widely affect the results generated and lead to inaccurate results.

Putting these aside and assuming the BERTScores are accurate, there exists disparities among scores between variables. The F1 scores range from 0.367 for the "Defendant" to 0.655 for the "Case Number". The second lowest scoring variable was "Serve Date". As seen in Figure 4.0.2

and Figure 4.0.2, the word choice and ordering change the output and could be the reason for such difference in F1 scoring. With a score of 0.367 at face value and running on a 0 to 1 scale, the model output and human verified response were not semantically similar. This again highlights issues with the RAG model, and or the scoring mechanism. The exact reasons for this is currently unknown and requires future study. Possible reasons for low scoring may include the prompting, unusual characters, or the model setup itself. Although the model overall shows mediocre similarity, individual questions will need to be checked before being put into use elsewhere.

5.2 Future implications of field

Creating datasets is the baseline for many processes implemented in current day that base decisions on data. By developing a dataset of court cases, it adds to the minuscule amount of legal data that is already published to train models. In this effort, more machine learning and artificial intelligence models can be built solely with the purpose of legal aid. This work also invites future academic work to study pro se cases as a whole. In the current climate of high gas prices, high cost of living, and other controlled and uncontrollable factors, pro se case data could be leveraged to better serve people who file this way. Especially in an environment where generative AI is already being in use, it can be built to serve communities that represent themselves and traverse the legal system. Even to build models such as the RAG built here, there are special model bases that are trained solely on legal documents due to the dense language and interpretations given in legal proceedings. Moving forward to usage of this dataset will be where it proves useful or not useful. More data may need to be collected in order to make useful future predictive modeling for court cases. This may include more variables such as a copy of legal standards at the time, influential legal cases, implementing generation in RAG, and more that can not be identified by the current case information alone. These variables to add and look for are also dependent on the model that is designed with this data. Based on the machine learning model of choice, the data found and added may need to be transformed into formats

accepted by the model itself. This lends hand in hand with the models and finetuning of a RAG system for retrieval. Such advances in this framework without generation would prove useful in dynamically growing the dataset and in line with a future application for pro se filers.

6
Future Work

- Safe guards for receiving unsuccessful response.status_codes should be put in place to safely and holistically scrape sites and request API's. Rate limiting steps, such as constantly putting the scripts to sleep, were introduced to not overload the API request systems, but does not account for the initial error in receiving a page. Codes such as 429 could mean too many requests in a given time frame, requests during site construction, or other errors in accessing a website, causing the rest of the scrape to fail or become incomplete.

- Scripts should be made more dynamic by adding single input cases when sites are updated, instead of forcing a total and clean scrape or API call over the whole website. To achieve this, implement webhooks where possible or other methods. This would allow for a growing list of cases and chronologies, and would extend the reach of cases past limited query results in dynamically generated websites. It also keeps the dataset up to date on new documents and case statuses. It is less efficient to keep the code base in its current standing and repetitively go over the whole website each time necessary. The RAG model can implement this change but would still require a check of what new information was added, and to download them locally to be compatible with this model to run.

- Currently, the processes do not account for cases that are repeated within the same source, or across varying sources. Cases that do appear multiple times across a source or between

sources should be analyzed and properly combined. This will provide insight into the chronology of the cases or if actual differences present themselves within the case versions. It should be examined why one receives multiple cases to begin with. Possibilities include new documentation added, a different version of the case, or a different court that it could be filed in. These discrepancies should be examined and merged into the chronology or other appropriate variables. After merging, searching for the case could be done with binary search and chronology could be held in a linked list or other data structure for ordering purposes.

- The RAG system itself should be optimized and corrected to a higher degree so that proper responses can be made. This may include testing different models, fixing referencing issues, as well as changing parameters. These changed parameters could include the chunking strategy or making a more distinct pipeline for the model to operate under. This should also handle chained chronological output to return a list of possible answers, given that this list is necessary for the variable type.

- Depending on implementation of the dataset, more variables may need to be extracted and added. This would ensure that models built upon the data have adequate points to analyze. Finding more sources of data could support this by providing more information on each case.

7
Limitations

7.1 Web Scraping

Web scraping is limited to the robots.txt file of each site. Many large search engines like Google, Bing, and Safari disallow scraper searching or limit searches to a minuscule amount to protect the sites and not overload their systems. This posed an issue when trying to find and access full case information or documents, and extracting information from Google Scholar. This is why cases from Google Scholar only have a link and no other information, as Google Scholar does not let scrapers traverse it according to the robots.txt. Having a free crawl of these engines would allow for more cases but would require a more abstract data scraping script to accommodate for different data formats.

The web scrapers themselves are either built by cherry picking HTML tags or generated with the JSON objects returned from API calls. With the structure of the current APIs and structure of websites, this limits how and what information can be extracted. Scraping directly from HTMLs is limited by the amount of dynamic data that the host allows to be viewed per page, which does not match the repertoire of data hosts have relative to the query. In development, some sites had format updates, rate limiting steps, or CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) bots that changed the

way that data could be collected through HTMLs. This is also a warning for future work, as this code base may not work for changed infrastructure of websites and API calls.

7.2 APIs

Also, not having access to certain API's forced an automated scraping of each query, which limits the amount and content of data that could be processed for this project. Particularly for usage of SerpAPI is the limited free monthly usage. After testing and debugging, running out of searches limits the amount of data one can access. Search queries include each paginated search which grows the totally usage. The process of acquiring the data is entirely upon the amount of queries already used, the amount of data to be collected, the number of cases presented in each search, and which plan one has for the platform. Under the free plan, users get 250 free monthly searches, followed by exponentially increasing amounts of searches and prices. To access the free enterprise, users must first sign up for the service with a validated email address and phone number.

The CourtListener scraper was built only with Python package "request" instead of the REST API set up and described on their home page. Unfamiliarity with the API structure and necessary language lead to utilization of the Python "requests" method, which limited the filtering options for JSONs that the REST API had more available. Also this method produced significantly less court cases than would be displayed in a live search with the same parameters.

7.3 RAG Model

The documents that were fed to the RAG were all homogeneous. They came from the same source, spanned relatively the same information, had some illegible plain text, and had the same format. To truly test and fine tune the model, it should be run across either more specific types of documents, or broaden the range to even more types and sources of documents. This will ensure the model is effective across varying platforms and not simply showing high success from this one source type but was not done due to time constraints.

The RAG system also can not account for a chained chronological ordering in its current state. It chooses the chunk with the highest logit score, but is not robust enough to select multiple answers necessary for chronology. For cases added through RAG, they will not have a chronology variable to operate under.

7.4 Lacking Multiple Instance Integration

This project lacks integration of cases already seen in previous steps for scripts not using RAG. Given a case that has been seen before, the models continue to add them as a rewritten instance to the datasets. What should happen is understand the ordering of the documents and add them to the earliest date of the same case's chronology. This would ensure unique cases in the dataset and a dynamic update of each case in current time. In the present state, a case could have multiple entries, leaving the chronology incomplete or requiring a more in depth explanation for why there are multiple versions of the same case.

8
Conclusion

The initial goal of this research was to curate the first large publicly available baseline for machine learning algorithms on pro se cases in Georgia. With the scraping and Retrieval Augmented Generation model built here, this goal has not successfully been met. The dataset has missing values and fragments of the larger understanding of cases. The scripts used to get the data could also be enhanced to account for varying types, multiple case integration, and can should improve efficiency. However, with the resources available, a sizable dataset and model were created that can take in mass amounts of documents at once with a high success rate. The RAG and BERTScore implementations for dynamic insertion were unsuccessful and require amending. Future work is paramount to making this dataset useful, improving the efficiency of code, and the accuracy of the RAG model. Until then, this dataset adds to the limited number of legal datasets, especially of the kind presented here through Georgia's civil pro se cases. This can also be scaled to include more geographies and data inputs to make a more robust data set. Between the work provided here and future work, there can be much more data driven decision making to help pro se filers within Georgia, and can be expanded to other states and countries thereafter.

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